Hierarchical Convexity-Run Decomposition of Sampled Curves: Finite-Sample Recovery and Stability

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Abstract. Many one-dimensional data sets are observed as sampled curves whose localized changes of convexity carry structural information not naturally described by a prescribed basis or frequency decomposition. We introduce Hierarchical Convexity-Run Decomposition (HCRD), a finite nonlinear transform that recursively replaces maximal runs of one discrete-curvature sign by endpoint chords. The construction produces nested knot sets, exact reconstruction, signed residual lobes, and linear total work in a sparse representation. We establish affine equivariance and variation control, characterize the global discontinuity of the hard knot map, and give finite-sample multilevel decision-margin certificates. For an alternating chord-lobe model observed under Gaussian noise, we prove recovery of all first-level convexity boundaries together with uniform baseline and detail error bounds, including approximate sampled joins. Simulations recover the predicted transition and delimit failure outside the certified regime. As a downstream test, we apply the representation to no-refit cross-study classification of expert-labelled chromatographic shapes. HCRD complements an established low-dimensional quality score in two transfers, while matched conventional controls and a larger laboratory shift identify regimes where superiority is not established. The results position HCRD as an interpretable geometric representation with explicit computational, recovery, and applicability boundaries.

Key words. discrete curvature, hierarchical representation, convexity change points, nonlinear transform, finite-sample recovery, sampled curves

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1. Introduction.

Many one-dimensional data sets are best viewed as sampled curves rather than as stationary time series. Their scientifically relevant structures may be localized convex or concave excursions, and the locations, signs, supports, and complete shapes of those excursions can matter more than a decomposition into prescribed frequency bands. We ask whether this geometry can be encoded by a finite representation that is adaptive to the samples, exactly reconstructing, computationally sparse, and accompanied by quantitative recovery statements under noise.

Hierarchical Convexity-Run Decomposition (HCRD) answers this question with a discrete operator. On an ordered grid, it partitions the polygonal graph into maximal runs of one divided-curvature sign, replaces each run by its endpoint chord, records the signed residual lobe, and recurses only on retained knots. The result is a nested tree of supports and residual shapes; it is a geometric hierarchy, not an orthogonal basis or a claim that levels are frequency bands. Figure 1 shows two steps and the exact synthesis identity.

The construction intersects three established lines of work but has a different estimand from each. Data-adaptive signal decompositions include empirical mode decomposition (EMD) [10], intrinsic time-scale decomposition (ITD) [8], iterative filtering [15], and convexity-driven geometric scale space [3]. ITD is the closest chord-baseline recursion, while the LULU discrete pulse transform is the closest exact nonlinear pulse hierarchy [1, 13]. These methods

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Table 1

Closest one-dimensional representations. "Exact" means that the native components and final baseline reconstruct the input.

Method	Primitive / estimand	Reconstruction / hierarchy	Computation and regularity
HCRD	Curvature-sign runs; endpoint-chord residual shapes	Exact; nested knot subsets	$O(n)$ total sparse work; hard decisions; irregular grid
ITD [8]	Extrema updates; baseline and rotations	Exact; recursive baseline	Linear work per level; hard extrema
LULU/DPT [1, 13]	Order / flat zones; constant pulses	Exact; size hierarchy	Linear Roadmaker algorithm; order based
RDP [6]	Maximum chord error; polygonal approximation	Exact with residual; split tree	Quadratic worst case; hard ties
Trend filtering [11, 18]	Penalized discrete derivatives; fitted trend	Exact with residual; no native component tree	Convex optimization; continuous fitted values
Piecewise-convex / QTA [17, 4]	Convexity episodes; fitted shape sequence	Statistical segmentation; not a decomposition	Constrained or mixed-integer optimization
Gaussian scale space [2]	Convolution scale; smoothed curves	Scale differences; scale ordered	Continuous in data; regular-grid standard

do not use the HCRD rule of maximal divided-curvature runs followed by nested eligible knot sets.

Shape-constrained estimation instead fits a latent curve subject to derivative sign restrictions. Trend filtering produces adaptively knotted piecewise polynomials through penalized discrete derivatives [11, 18]. Piecewise-convex estimation treats the number and locations of convexity changes as a model-selection problem [17]. Qualitative trend analysis identifies episodes with prescribed first- or second-derivative signs through shape constraints and mixed-integer model selection [4]. HCRD does not optimize a fitted curve or select a statistical model: it transforms the observed polygonal curve into an exact component hierarchy. Conversely, the hard HCRD knot map is only locally stable, whereas penalized estimators trade exact raw decisions for global regularity. The recent shape-constrained inference literature provides the broader statistical context for this distinction [7].

Multiscale change-point procedures provide simultaneous inference and sharp localization for piecewise-constant or approximately segmented signals [9, 14]. HCRD boundaries are changes in the sign of discrete curvature rather than jumps in a mean parameter. Our finite-sample results are therefore class-specific: they certify the exact branches visited by the hierarchy and recover convexity boundaries on an explicit chord-lobe model. They do not claim minimax separation from all adaptive estimators.

Table 1 summarizes these distinctions in estimand, reconstruction, computation, and regularity.

The paper also tests whether the representation carries useful information in a real cross-domain setting. Expert curation of pre-bounded LC–MS traces is appropriate because the endpoint chord is meaningful and signed compact-lobe morphology determines quality. This

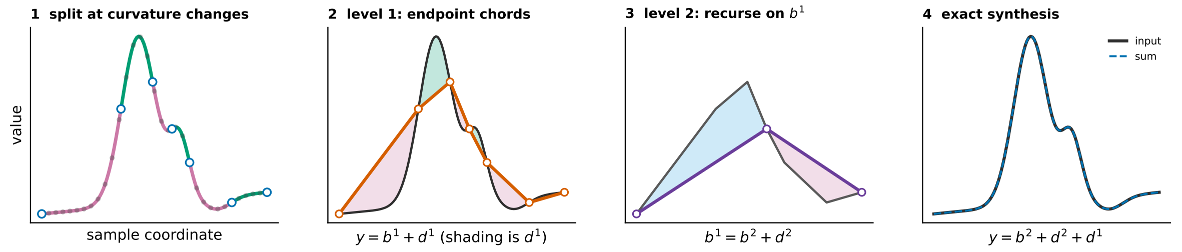

Figure 1. *Two HCRD steps on a generic sampled curve. Curvature-sign changes delimit retained knots. Replacing each level-1 run by its endpoint chord gives b^1 and signed residual d^1; recursing on b^1 gives b^2 and d^2. The displayed truncation reconstructs $y = b^2 + d^2 + d^1$ exactly. Shading shows complete residual shapes rather than frequency bands.*

application is used to test transfer and failure boundaries, not to define the method or its theory.

Our contributions are:

1. a complete finite curvature-run algorithm with exact reconstruction, nested knot sets, linear total sparse work, affine equivariance, variation control, and interpretable signed residual shapes;
2. finite-sample multilevel knot-tree agreement under explicit decision margins, together with boundary and component recovery on an alternating chord-lobe class with exact or approximate sampled joins;
3. a sharp obstruction to global continuity of the hard map and stable alternatives that make the regularity–interpretability tradeoff explicit; and
4. theorem-linked phase experiments, computational audits, and frozen cross-study validation with matched conventional controls and retained negative results that delimit the method's applicability.

The remainder of the paper defines the transform, establishes its structural and stability properties, derives the noise and recovery results, and then reports synthetic, computational, and cross-study evidence. Detailed proofs, inference extensions, and secondary experiments are provided in the supplement.

2. Definition of the transform.

Let $x_0 < \cdots < x_{n-1}$ and $y = (y_0, \ldots, y_{n-1}) \in \mathbb{R}^n$. Define divided slopes and discrete curvatures by

$$(2.1) \qquad s_i(y) = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}, \qquad \kappa_i(y) = s_i(y) - s_{i-1}(y).$$

For absolute and relative tolerances $a, r \geq 0$, define

$$(2.2) \qquad \tau = a + r \max\{1, \max_i |s_i(y)|\}.$$

A curvature is active when $|\kappa_i| > \tau$ and inactive otherwise. The mathematical hard operator uses $a = r = 0$; the implementation defaults to a tiny relative tolerance to suppress floating-point curvature on interpolated affine segments. Algorithm 2.1 specifies every branch of the centred rule.

Algorithm 2.1 One centred HCRD hierarchy on an irregular grid

Input: strictly increasing x, samples y, tolerances $a, r \geq 0$, and optional depth cap L. Initially $E = (0, \ldots, n-1)$.

1. On the current eligible list $E = (e_0, \ldots, e_{m-1})$, set $z_i = y_{e_i}$, $u_i = x_{e_i}$, compute divided curvatures, and compute τ from the current divided slopes. If $m = 2$, stop.
2. Set the local start $p = 0$ and the new local knot list $K = (0)$.
3. Let f be the first active curvature index in $\{p+1, \ldots, m-2\}$. If none exists, append $m-1$ to K and finish this level. Otherwise set the active sign to $\operatorname{sign}(\kappa_f)$, $\ell = f$, and scan right.
4. Ignore inactive curvatures. On each active curvature of the same sign, update ℓ. Let q be the first active curvature of the opposite sign. If no such q exists, append $m-1$ and finish the level.
5. If $q - \ell = 2$, the single inactive curvature at $\ell + 1$ is the sampled transition and the candidate is $c = \ell + 1$. At zero tolerance this is exactly the sampled-zero case.
6. Otherwise set $c = \ell$ only when $|\kappa_\ell| < |\kappa_q|$ and $\ell \neq f$; set $c = q$ in every other case. Thus equal magnitudes are tied to the right, first-opposite side, and a long inactive plateau never creates an interior knot.
7. Enforce strict coarsening by replacing c with $\max\{p+2, \min(c, m-1)\}$. Append c to K, set $p = c$, and return to step 3. The final remainder always closes at $m-1$.
8. Map the local knots back to original indices, $E' = (e_k : k \in K)$. Interpolate the endpoint chords through (x_i, y_i) for $i \in E'$ to obtain the next baseline; subtract it to obtain the signed detail.
9. Set $E \leftarrow E'$ and repeat until $|E| = 2$ or the optional cap L is reached. Later levels therefore use only previously retained knots.

Let $K(y)$ denote the first mapped knot set and let Ty be linear interpolation through (x_k, y_k) for $k \in K(y)$.

Set $b^0 = y$ and recursively define

$$b^{j+1} = Tb^j, \qquad d^{j+1} = b^j - b^{j+1}. \tag{2.3}$$

The implementation restricts eligible knots at level $j+1$ to knots retained at level j. This prevents the insertion of geometrically redundant samples inside an existing affine segment. In local eligible-knot coordinates, a nonterminal centred interval is required to advance by at least two old intervals; the first-opposite and sampled-zero cases below satisfy this convention explicitly.

3. Structural results.

Theorem 3.1 (Reconstruction, nesting, and sparse complexity). *For every computed level J, the transform satisfies the following three properties.*

(i) Exact reconstruction: $y = \sum_{j=1}^{J} d^j + b^J$.

(ii) Finite nesting: *the eligible knot sets are nested, and every nonterminal HCRD interval consumes at least two intervals of the preceding level. The hierarchy therefore ends at the endpoint chord after at most* $\max\{1, \lceil \log_2(n-1) \rceil\}$ *implemented levels.*

(iii) Sparse linear work: *if $N_0 = n-1$ and N_j is the number of intervals after level j, a knot-only implementation performs $O(n)$ total knot-walk work and stores*

$$\sum_{j=1}^{J} |K_j| < 2(n-1) + J = O(n). \tag{3.1}$$

Dense baselines and details can be reconstructed exactly on demand.

Proof. Substitution of $d^j = b^{j-1} - b^j$ telescopes, proving (i). Eligibility gives nesting. A run can close only after the walk has passed its first active old curvature and reached either an eligible intervening zero or the first active curvature of opposite sign. Thus an old interior knot disappears in every nonterminal new interval, which gives the halving recurrence and (ii). In particular, $N_j \le \lceil N_0/2^j \rceil$ and $J = O(\log n)$. Since $\lceil N_0/2^j \rceil \le (N_0 - 1)/2^j + 1$, summing the interval and knot counts proves (iii). Chord interpolation through each stored knot set recovers the dense baseline, and adjacent differences recover the details. ∎

Proposition 3.2 (Signed structures). *On a discretely convex run, $y - Ty \le 0$. On a discretely concave run, $y - Ty \ge 0$.*

Proof. A convex polygonal graph lies below its endpoint secant. Reverse the inequality for a concave graph. ∎

For structure r of detail d^j on $I_{jr} = [x_a, x_b]$, define its signed area, geometric mass, and quadratic residual energy by

$$S_{jr} = \int_{I_{jr}} d^j(x)\,dx, \qquad A_{jr} = \int_{I_{jr}} |d^j(x)|\,dx, \qquad E_{jr} = \int_{I_{jr}} d^j(x)^2\,dx, \tag{3.2}$$

where d^j is the exact piecewise-linear interpolant of the sampled residual. These quantities are computable from retained knots without materializing a length-n detail. Signed structures give $A_{jr} = |S_{jr}|$. Under $y \mapsto cy + \alpha + \beta x$, A_{jr} scales by $|c|$ and E_{jr} by c^2; under an orientation-preserving horizontal affine change $x \mapsto \rho x + \tau$, both integrals acquire the Jacobian factor ρ. Thus A_{jr} is a literal morphology measure in signal-units times coordinate-units. It is not spectral energy, and the nonorthogonal hierarchy has no Parseval identity.

Proposition 3.3 (Dimensionless lobe shape). *For a nonzero structure let $W = x_b - x_a$, $H = \max_{I_{jr}} |d^j|$, and define*

$$F = \frac{A_{jr}}{WH}, \qquad C = \frac{A_{jr}^2}{WE_{jr}}. \tag{3.3}$$

Then $0 < F \le 1$ and $0 < C \le 1$. Both are invariant under nonzero vertical scaling, affine trend addition, and orientation-preserving horizontal affine changes. A piecewise-linear triangular lobe has $(F, C) = (1/2, 3/4)$ regardless of apex location; a parabolic lobe $4Ht(W - t)/W^2$ has $(F, C) = (2/3, 5/6)$.

Proof. $A \le WH$ follows from $|d| \le H$, while Cauchy–Schwarz gives $A^2 \le WE$. The scaling laws above cancel in both ratios. Direct integration gives the two examples. ∎

The first ratio is the implemented per-structure shape factor. The second is available as a quadratic concentration descriptor but was not added after the LC–MS protocols were frozen. Keeping A_{jr} or either invariant for each moving support and level also yields a low-dimensional temporal morphology series; collapsing all structures to one total mass discards that evolution.

Theorem 3.4 (Affine equivariance). *For zero curvature tolerance, nonzero a, and $\ell_i = \alpha + \beta x_i$,*

$$T(ay + \ell) = aTy + \ell. \tag{3.4}$$

Hence every HCRD detail is invariant to affine trend addition and scales by a.

Proof. Affine addition cancels from adjacent divided-slope differences. Multiplication by a preserves all transition locations, although negative a reverses the signs. Chord interpolation commutes with both operations. ■

Theorem 3.5 (Range and total variation). *For one HCRD step,*

$$\min y \leq (Ty)_i \leq \max y, \qquad TV(Ty) \leq TV(y). \tag{3.5}$$

Proof. Every chord sample is a convex combination of its endpoint values. Its variation equals the absolute difference between the endpoint values, which is no greater than the variation of the removed polygonal subpath. ■

4. Stability and its obstruction.

Theorem 4.1 (Stability inside a curvature-decision cell). *On a unit grid suppose $\gamma = \min_i |\Delta^2 y_i| > 0$ and let*

$$\eta = \min_{\Delta^2 y_i \Delta^2 y_{i+1} < 0} \big||\Delta^2 y_i| - |\Delta^2 y_{i+1}|\big|, \tag{4.1}$$

with $\eta = +\infty$ if there is no sign transition. For the centred rule, if $\|e\|_\infty < \min\{\gamma/4, \eta/8\}$, then $K(y+e) = K(y)$ and

$$\|T(y+e) - Ty\|_\infty \leq \|e\|_\infty, \tag{4.2}$$

$$\|(I-T)(y+e) - (I-T)y\|_\infty \leq 2\|e\|_\infty. \tag{4.3}$$

Proof. The coefficient norm of the second-difference stencil $(1, -2, 1)$ is four, so the first margin prevents a curvature-sign change. A difference of two absolute curvature magnitudes moves by at most $8\|e\|_\infty$, so the second margin preserves every centred transition comparison. The knot walk is therefore fixed. Fixed-knot interpolation is a convex combination of endpoint perturbations, and the detail estimate follows by the triangle inequality. ■

The second margin is necessary: $(0, 0, 1, 3, 4)$ has curvatures $(1, 1, -1)$, whereas $(0, 0, 1, 3-\varepsilon, 4-2\varepsilon)$ has the same signs but selects the other side of the tied centred transition. The legacy first-opposite rule does not compare magnitudes and requires only the sign margin.

Proposition 4.2 (Global discontinuity). *The raw HCRD baseline operator is not globally continuous.*

Proof. For $y^\pm = (-2, 0, 2 \pm \varepsilon, -2)$, the inputs approach one another as $\varepsilon \downarrow 0$, whereas one first baseline retains the interior peak and the other becomes the constant endpoint chord. Their sup-norm separation tends to four. ■

Theorem 4.3 (No continuous generic-agreement extension). *For four-sample signals let $G = \{y : \min_i |\Delta^2 y_i| > 0\}$. There is no globally continuous map $F : \mathbb{R}^4 \to \mathbb{R}^4$ that agrees with the raw HCRD baseline T on all of G.*

Proof. For every $0 < \varepsilon < 1$, both $y_\varepsilon^\pm = (-2, 0, 2 \pm \varepsilon, -2)$ lie in G and converge to the same y^0. Agreement on G makes $F(y_\varepsilon^\pm)$ equal the two HCRD outputs in the preceding counterexample, whose limits are separated by four. Continuity at y^0 would force both limits to equal $F(y^0)$, a contradiction. ∎

Remark 4.4. HCRD is not an orthogonal transform and need not contract Euclidean energy. For $y = (-2, -1, -2)$, one has $Ty = (-2, -2, -2)$ and $\|Ty\|_2 > \|y\|_2$.

5. Stable companion representations.

Hard knot decisions are locally stable inside a curvature-decision cell but cannot be made globally continuous. Two complementary summaries retain interpretability while giving global perturbation control. First, for the divided-curvature matrix D and a proper closed convex function ϕ, define

$$P_\phi y = \operatorname*{argmin}_z \left\{ \frac{1}{2}\|y - z\|_2^2 + \phi(Dz) \right\}, \qquad Q_\phi = I - P_\phi. \tag{5.1}$$

The standard proximal-map result makes both P_ϕ and Q_ϕ firmly nonexpansive, gives the exact split $y = P_\phi y + Q_\phi y$, and makes the split equivariant under affine additions in $\ker D$ [16]. Thus it supplies a stable outer guide without changing the hard hierarchy into a new convex estimator.

Second, signed zero-dimensional superlevel persistence of $u_y^+ = (D_x y)_+$ and $u_y^- = (-D_x y)_+$ gives a stable lobe summary. Classical persistence stability yields

$$d_{\mathrm{SC}}(y, z) \le C_x \|y - z\|_\infty, \qquad C_x = 2 \max_i \{h_{i-1}^{-1} + h_i^{-1}\}, \tag{5.2}$$

and therefore protects every finite signed-curvature bar whose lifetime is greater than $2C_x\|y - z\|_\infty$ [5].

Proposition 5.1 (Curvature visibility limit). *If $y(t) = g(t) + a\sin(\omega t)$ and $g''(t) > |a|\omega^2$ on an interval, then y'' has one sign there; a raw inflection-run rule cannot isolate the oscillation on that interval.*

Complete definitions, proofs, and numerical stability audits are given in the supplement.

6. Noise-aware hierarchy and recovery guarantees.

6.1. Simultaneous curvature control and multilevel agreement.

Assume a uniform grid with spacing h and observations $Y_i = y_i + \varepsilon_i$, where the errors are independent $N(0, \sigma^2)$ variables.

Theorem 6.1 (Family-wise curvature threshold). *For $0 < \delta < 1$, let*

$$\tau = \frac{\sigma}{h}\sqrt{12 \log \frac{2(n-2)}{\delta}}. \tag{6.1}$$

With probability at least $1 - \delta$, every observed curvature differs from its population value by at most τ. On this event, population zero curvatures are thresholded to zero and all curvatures with magnitude greater than 2τ retain their correct active sign.

Proof. Each curvature error has variance $6\sigma^2/h^2$. Apply a Gaussian tail bound to each of the $n-2$ entries and take a union bound; independence of curvature errors is not required. The classification statement follows from the triangle inequality. ■

For a common physical-unit alternative to the row-normalized anomaly score, let $h_{\min} = \min_i(x_{i+1} - x_i)$,

$$\epsilon = \sigma\sqrt{2\log(2n/\delta)}, \qquad \eta = 4\epsilon/h_{\min}, \qquad \tau = \lambda\eta, \quad \lambda \geq 1, \tag{6.2}$$

run HCRD with absolute curvature tolerance τ, and set

$$Z(t) = \Big[\max_j |d_j(t)| - 2\epsilon\Big]_+ . \tag{6.3}$$

Theorem 6.2 (Multilevel agreement margins). *Run the noiseless minimum-curvature hierarchy with tolerance τ through level L. At every visited level suppose each curvature declared inactive satisfies $|\kappa_i| + \eta \leq \tau$, each active curvature satisfies $|\kappa_i| - \eta > \tau$, and each comparison $|\kappa_i| < |\kappa_j|$ used to choose a side of an unsampled sign transition has gap $\big||\kappa_i| - |\kappa_j|\big| > 2\eta$. Then, with probability at least $1-\delta$, the noisy and noiseless knot sets agree through level L. Consequently a retained noiseless chord lobe of height $H > 4\epsilon$ has strictly positive score at its noiseless peak on that event.*

Proof. On $\|e\|_\infty \leq \epsilon$, every active-grid curvature moves by at most η. The first two margins preserve threshold status and sign; the final margin preserves every magnitude comparison. Thus the complete first-level branch trace is unchanged. Induction applies because equal knot sets expose the same retained grid at the next level. Under agreement, a density changes by at most 2ϵ; subtracting the null radius gives the peak lower bound $[H - 4\epsilon]_+$. ■

6.2. Generative recovery on a chord-lobe class. On a uniform grid $x_i = x_0 + ih$, define the divided-slope curvature

$$\kappa_i(z) = \frac{z_{i+1} - 2z_i + z_{i-1}}{h}, \qquad 1 \leq i \leq n-2. \tag{6.4}$$

This scaling matches Algorithm 2.1; it is a slope difference rather than a divided second derivative.

Theorem 6.3 (Finite-sample generative chord-lobe recovery). *Let $0 = k_0 < k_1 < \cdots < k_J = n-1$, with $k_r - k_{r-1} \geq 2$, and write $f = b + q$. Suppose b is affine on each $[x_{k_{r-1}}, x_{k_r}]$, $q_{k_r} = 0$, and the population curvatures of f satisfy*

1. *$\kappa_{k_r}(f) = 0$ at every interior knot;*
2. *inside successive knot intervals they have alternating constant sign and magnitude at least $\gamma > 0$.*

Observe $Y_i = f_i + e_i$ with independent $e_i \sim N(0, \sigma^2)$ and fix $0 < \delta < 1$. Set

$$\tau_\delta = \frac{\sigma}{h}\sqrt{12\log\frac{4(n-2)}{\delta}}, \qquad \epsilon_\delta = \sigma\sqrt{2\log\frac{4n}{\delta}}. \tag{6.5}$$

Run the first centred HCRD step with absolute curvature tolerance τ_δ. If $\gamma > 2\tau_\delta$, then with probability at least $1-\delta$ its knot set is exactly $\{k_0, \ldots, k_J\}$. On the same event, for its chord baseline $\widehat{b}$ and detail $\widehat{q} = Y - \widehat{b}$,

$$\|\widehat{b} - b\|_\infty \leq \epsilon_\delta, \quad n^{-1}\|\widehat{b} - b\|_2^2 \leq \epsilon_\delta^2, \quad \|\widehat{q} - q\|_\infty \leq 2\epsilon_\delta, \quad n^{-1}\|\widehat{q} - q\|_2^2 \leq 4\epsilon_\delta^2. \tag{6.6}$$

Proof. With probability at least $1-\delta/2$, every curvature error is at most τ_δ: this is Theorem 6.1 with its error budget replaced by $\delta/2$. Population zeros are then inactive, whereas $\gamma > 2\tau_\delta$ makes every active curvature remain active with the correct sign. Hence each declared knot is the unique inactive eligible point between opposite active blocks. The centred transition rule selects that point, and the two-segment gap prevents the strict-coarsening correction from moving it. Induction along the first-level walk proves exact knot recovery.

Independently, a Gaussian maximum bound gives $\|e\|_\infty \le \epsilon_\delta$ with probability at least $1-\delta/2$. On exact recovery, $f_{k_r} = b_{k_r}$ and b is its own chord interpolant, so $\widehat{b} - b$ is the chord interpolant of the knot errors. Its sup norm is at most $\|e\|_\infty$; subtracting it from e gives the detail bound. The MSE bounds follow from the sup-norm bounds, and a union bound completes the proof. ∎

Corollary 6.4 (Recovery with approximate sampled joins). *Retain the model, knot spacing, endpoint, active-block, and Gaussian-noise assumptions of Theorem 6.3, but replace its zero-join condition by the fixed structural bound*

$$|\kappa_{k_r}(f)| \le \eta, \qquad 1 \le r < J, \quad \eta \ge 0. \tag{6.7}$$

Run the first centred HCRD step with relative tolerance zero and any absolute tolerance t satisfying

$$\tau_\delta \le t \le \eta + \tau_\delta. \tag{6.8}$$

This interval includes both the original noise-only tolerance $t = \tau_\delta$, which does not require knowledge of η, and the join-inactivating tolerance $t = \eta + \tau_\delta$. If

$$\gamma > \eta + 2\tau_\delta, \tag{6.9}$$

then, with probability at least $1-\delta$, the knot set is exactly $\{k_0, \ldots, k_J\}$. The four baseline and detail bounds in Theorem 6.3 hold on the same event.

Proof. On the simultaneous curvature event from Theorem 6.3, each observed join has magnitude at most $\eta + \tau_\delta$, whereas every adjacent active curvature has magnitude at least $\gamma - \tau_\delta > \eta + \tau_\delta$, remains above t, and retains its population sign. If a join is inactive at t, the isolated-inactive rule selects it. If it is active, its sign agrees with either the left or right block. The sign transition then occurs immediately after or before the join; in both cases the centred minimum-magnitude rule selects the join because it is strictly smaller than the adjacent active curvature. The two-segment gap keeps the smaller left-side candidate past the first active curvature. Thus every declared knot is recovered. The sample-maximum event and interpolation argument are unchanged. ∎

The bound η is fixed independently of the observed noise when the corollary is used as a certificate. The noise-only implementation itself does not estimate or use η; a data-derived certificate would require a simultaneous upper confidence bound.

The assumptions are structural rather than circular: they describe sampled curvature before running HCRD. An explicit subclass has $k_r = rm$, a globally affine b, and alternating parabolic lobes with amplitudes $A_r > 0$,

$$q_i = s(-1)^r 4A_r u_i(1-u_i), \qquad u_i = (i - rm)/m, \quad rm \le i \le (r+1)m. \tag{6.10}$$

Here

$$\gamma = \frac{8 \min_r A_r}{m^2 h}, \qquad \eta = \frac{4(m-1)\max_r |A_r - A_{r-1}|}{m^2 h}. \tag{6.11}$$

Equal amplitudes give $\eta = 0$ and recover the theorem's exact sampled joins; variable amplitudes fall under Corollary 6.4 whenever its separation condition holds. When that separation fails, unequal adjacent amplitudes can move a boundary by one sample even without noise. The guarantees are sufficient, not necessary, separation conditions.

Optional inference extensions.. Fixed or independently guided HCRD lobes can also be used as templates in finite or continuous scans, and independent replicates or buffered folds can separate structure selection from testing. Matching-order dictionary bounds, projected-norm calibration, replicate pivots, and nested-tree gatekeeping are developed in the supplement. They are optional inferential uses of HCRD rather than assumptions needed by the transform or recovery results below.

7. Experiments.

The primary comparison protocol was frozen locally before inspecting results; it was not deposited in a public preregistration registry. It tests affine chord baselines with alternating convex/concave equal-amplitude parabolic lobes. Generic smoother hyperparameters are selected by an oracle against the known latent baseline separately for each trial. The primary endpoint is baseline mean squared error. A task-specific superiority statement requires a paired bootstrap interval below zero and a multiplicity-corrected exact sign test. Out-of-class tests include polynomial trends, crossing chirps, weak oscillations under strong background curvature, boundary outliers, irregular sampling, and non-Gaussian noise.

The theorem-linked R1 experiment varied $\rho = \gamma/\tau_\delta$ over 11 values on four fixed event-count/width configurations: $(J, m) = (2, 4), (4, 8), (8, 16), (16, 8)$. Each cell contained 1000 independent draws, for 44,000 trials in total. Figure 2 shows a clear transition. Exact recovery ranged from 0.433 to 0.879 across configurations at $\rho = 1.45$, from 0.980 to 0.991 at $\rho = 1.75$, and was 0.999 throughout at $\rho = 1.90$. In all eight cells above the strict theorem boundary $\rho > 2$, it was 1.000; the smallest cellwise 95% Wilson lower bound was 0.9962. The joint event of exact knots and both reconstruction bounds had minimum probability 0.9980 above the boundary (minimum Wilson lower bound 0.9927). The result validates a sufficient condition on its declared class, not a necessary boundary for arbitrary waveforms.

Figure 3 maps departure from exact sampled joins. The corresponding experiment varied both η/τ_δ and γ/τ_δ over 162,000 independent signals with unequal adjacent lobe amplitudes. We compared the noise-only tolerance $t = \tau_\delta$ with the conservative join-inactivating choice $t = \eta + \tau_\delta$. All 72,000 draws in the strict certified region $\gamma/\tau_\delta > \eta/\tau_\delta + 2$ recovered every boundary for both choices (smallest cellwise 95% Wilson lower bound 0.9962), with no implication violation. Outside that region the noise-only choice was never worse cellwise and was often markedly better, confirming that η is useful for a certificate but need not be supplied to the algorithm.

7.1. Synthetic comparisons and computational audit.

On the exact chord-lobe class, centred HCRD recovered the latent chord baseline exactly and outperformed oracle-tuned EMD, VMD, ITD, trend filtering, CEEMDAN, and Iterative Filtering baselines. This class-matched result illustrates the geometric distinction rather than global dominance over mode

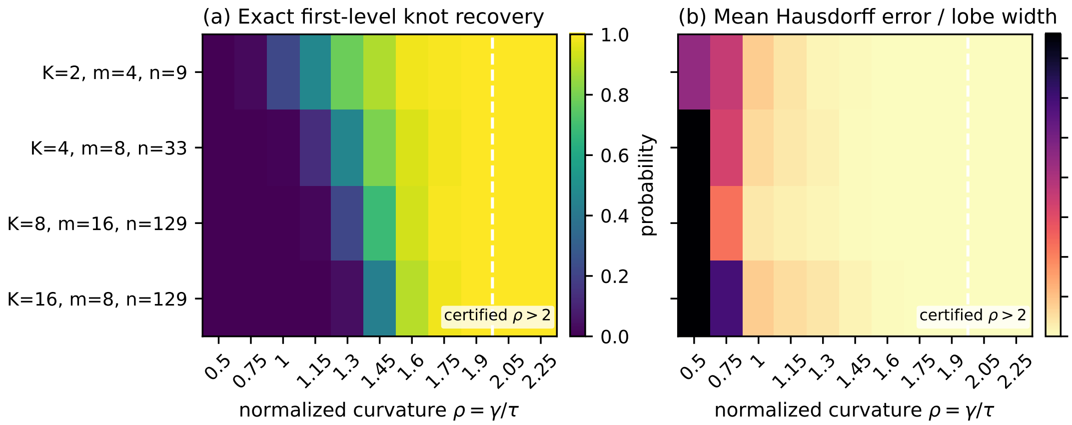

Figure 2. *Theorem-linked recovery phase diagram under iid Gaussian noise. Rows vary lobe count J, samples per lobe m, and total length n; columns vary normalized active curvature $\rho = \gamma/\tau_\delta$. Left: exact first-level knot recovery. Right: symmetric Hausdorff boundary error divided by lobe width. The dashed line marks the strict sufficient region $\rho > 2$; all 8000 draws in that region recovered the exact knots.*

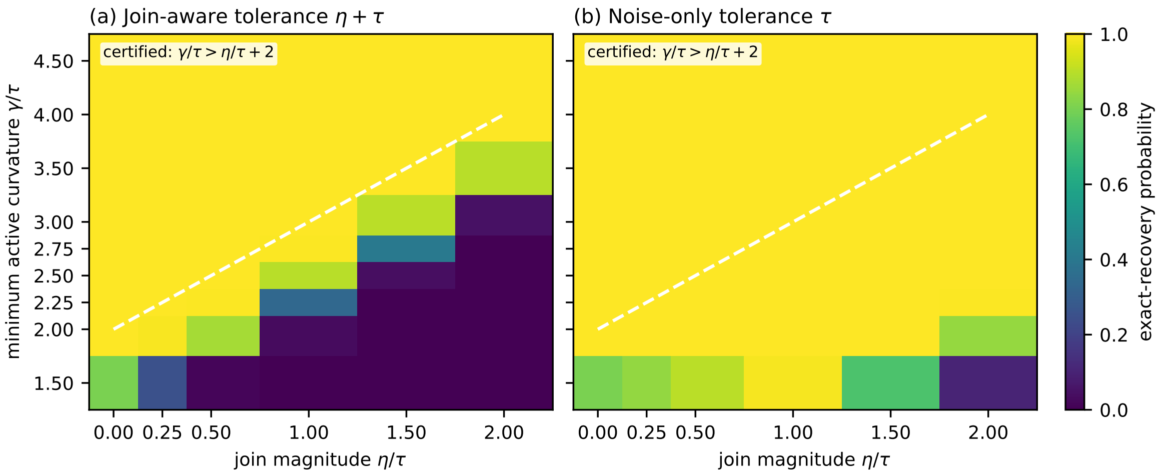

Figure 3. *Departure from exact sampled joins. Exact first-level recovery is shown against active curvature and join departure, both normalized by the simultaneous noise threshold. The dashed line is the strict sufficient boundary $\gamma/\tau_\delta > \eta/\tau_\delta + 2$. Left: the conservative join-inactivating tolerance. Right: the original noise-only tolerance, which does not require knowledge of η.*

decompositions. With Gaussian noise, a plug-in MAD-thresholded variant retained lower mean baseline error than the same comparators in fresh-seed confirmations. Sparse knot-only storage preserved downstream features bitwise, reduced construction time 11.89-fold relative to dense materialisation, and independent-signal process parallelism reached 1.96-fold throughput speedup on 3840 signals. Complete settings, paired intervals, tables, stable-guide experiments, and the CWRU ceiling result are reported in the supplement.

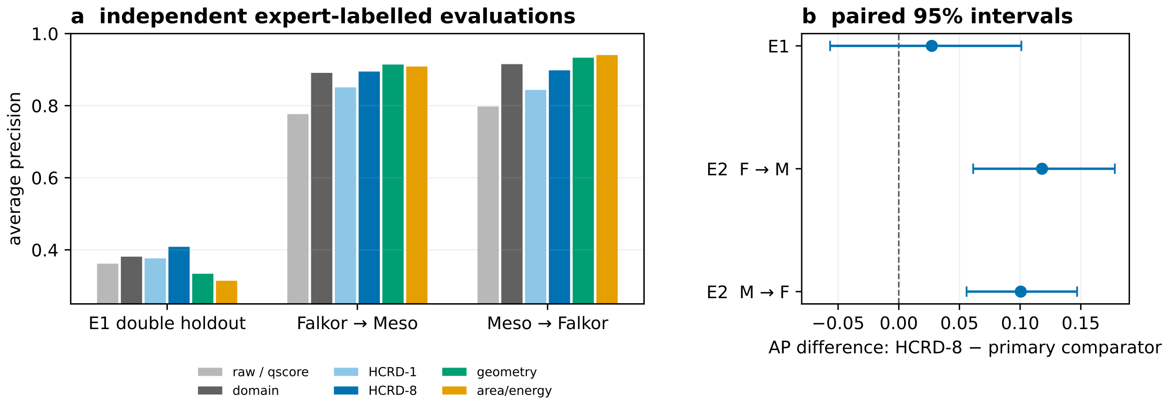

Figure 4. *Independent expert-labelled LC–MS transfer evidence. Panel a shows average precision for the preliminary double holdout and both frozen no-refit transfers. Panel b gives the predeclared HCRD-8 improvement over the primary comparator; E2 intervals are conditional on the fitted source models.*

7.2. Chromatographic transfer study. We used expert-labelled liquid-chromatography mass-spectrometry waveforms only as a downstream transfer test of the generic representation. The primary design was frozen before waveform inspection: one standardized logistic learner was trained on 1821 unambiguous Falkor features and applied without refitting to 2193 MESOSCOPE features, followed by the reverse direction. The minimal comparator was the published two-component quality score of Kumler et al. [12]; details of its independent reconstruction and the preliminary double holdout are given in the supplement.

Figure 4 summarizes the preliminary holdout and the two frozen no-refit transfer directions. Adding the eight-level HCRD representation increased average precision from 0.7774 to 0.8955 and from 0.7984 to 0.8990. The paired improvements were 0.1181 ($[0.0613, 0.1781]$) and 0.1005 ($[0.0559, 0.1470]$), with Holm-adjusted $p = 0.00040$ in both directions. All eight contrasts obtained from four predeclared implementations of the minimal score had positive 95% intervals. Multilevel HCRD also had positive point differences from the one-level representation in both directions, although only one survived multiplicity correction.

Two sensitivity analyses delimit that result. Against an equal-width Gaussian-derivative feature bank, the HCRD difference was -0.0108 ($[-0.0594, 0.0405]$) in one direction and 0.1175 ($[0.0624, 0.1654]$) in the reverse direction. Fixed-source 60-second target-block intervals remained positive, whereas intervals that also resampled and refit the source model crossed zero. The evidence therefore supports a conditional transfer gain over the compact published score, not uniform superiority over every comparably rich representation.

External application boundaries.. On the conditionally labelled Pttime subset, HCRD-8+Q achieved residual-error AP 0.5085 versus 0.0391 for qscore and retrieved 7 of 17 bad cases within the top 5% review queue, compared with none for qscore. This is a conditional reranking result, not a full-cohort estimate. Under the larger TARDIS/FAME laboratory shift, the HCRD-8+Q gain over qscore was 0.0725 but its interval $[-0.0155, 0.1345]$ crossed zero; HCRD-8 also tied HCRD-1 and a compact conventional bank was point-best. Additional

LC–MS, ECG, PPG, anomaly, classification, and prognostic boundary studies are reported in the supplement. Collectively they delimit the supported setting to pre-bounded compact-lobe curation rather than generic signal decomposition.

8. Discussion.

HCRD is a geometric multiscale representation of sampled curves, not a frequency decomposition. Its estimand is explicit: maximal runs of one-sign divided curvature are replaced by endpoint chords, and successive residuals encode signed departures from those chords. The construction is exactly reconstructing, terminates in logarithmic depth, and admits linear total work when only retained knots are stored.

The recovery theory separates what is global from what is conditional. The hard knot map is necessarily discontinuous at branch boundaries, while the proximal split and signed-curvature persistence give globally stable companion summaries. Away from those boundaries, deterministic decision margins imply agreement of the full hierarchy through any fixed depth. On the declared alternating chord-lobe class, those margins further yield exact first-level boundary recovery and explicit component-error bounds. This is a finite-sample mechanism result, not a minimax comparison with all adaptive dictionaries.

The empirical results are consistent with this scope. Class-matched synthetic experiments verify the predicted recovery regime and the sparse implementation preserves the dense representation exactly. In the expert-labelled LC–MS case study, HCRD adds transferable information to a compact published score, but an equal-width conventional bank ties it in one direction and source-refit block intervals include zero. Application-specific studies in the supplement also identify settings in which established detectors or shape priors remain preferable.

The method is therefore most plausible when a curve is already bounded, its endpoint chord is a meaningful local reference, and relevant structure appears as signed convex or concave excursions. It is poorly matched to instantaneous frequency estimation, oscillations hidden by stronger background curvature, or unthresholded samplewise decisions under heavy noise. Future work should estimate noise and dependence parameters from data, quantify localisation for unsampled transitions, and study broader polygonal generative classes and independent application cohorts.

Code and data availability. Software and reproducibility materials are archived on Zenodo (doi:10.5281/zenodo.22171977); active development is on GitHub. Large caches are rebuilt locally. Code: MIT; original research materials: CC BY 4.0.

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